

Extended Profiles Including Surface-Layer Wind Speed Improve Farm-Scale Wind Power Forecasting

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Key Points:

- Farm-scale wind power is better represented by vertical wind profiles extending below the rotor-swept layer.
- Under free-stream flow, full vertical profile matters, whereas under wake-influenced flow surface wind indicates profile intensity.
- Extended profiles improve power reconstruction, and a wind-direction-aware strategy enhances day-ahead forecasting.

Abstract

Wind farm power production reflects atmospheric variability across height and time scale, yet most forecasting approaches rely on wind speed at a single hub height. While rotor-equivalent wind speed accounts for variability within the rotor-swept layer, observations from a flat-terrain wind farm show that the profile relevant to farm-scale power extends below this layer. Partitioning observations by inflow regime reveals distinct mechanisms: under free-stream conditions, 10-m and 70-m winds provide complementary information for characterizing vertical structure, whereas under wake-influenced conditions, 10-m wind represents inflow profile intensity. This effectiveness arises from strong vertical coherence at synoptic time scales, enabling 10-m wind to characterize overall profile intensity across regimes, while rotor-swept layer winds become less reliable under wake-influenced conditions. An extended profile representation demonstrates consistent gains in power reconstruction and NWP-driven day-ahead forecasting. These results clarify how vertical wind structure governs farm-scale power and inform wind-to-power representation in operational forecasting.

Plain Language Summary

Wind turbines generate electricity from wind blowing across their blades, so knowing how wind varies with height is essential for predicting wind farm power. Most forecasts rely on wind measured at a single height near the middle of the turbine rotor, but this can miss important information. When turbine wakes disturb winds at rotor height, near-surface winds remain largely unaffected and better reflect the broader atmospheric conditions that control how much power the farm can produce. This is because winds at different heights tend to strengthen and weaken together over the hours-to-days time scales most relevant for power generation. By combining wind measurements from near the surface and the rotor layer, we improved both power estimates based on observations and day-ahead power forecasts. These findings suggest that existing weather towers and surface observations near wind farms can be used more effectively, and may also help inform how observational networks are designed and used for wind energy applications.

1 Introduction

Accurate day-ahead wind power forecasting is essential for power-system operation, electricity market, and renewable energy integration (Holttinen, 2005; Mahoney et al., 2012). Forecasting performance has been substantially advanced in recent years, including through the development of deep learning models (Liu & Zhang, 2024), and depends not only on model capability but fundamentally on the physical relevance of meteorological inputs (Optis & Perr-Sauer, 2019). Previous research has largely focused on model development, including architecture innovation (Bentsen et al., 2023), hyperparameter optimization (Hanifi et al., 2024), and post-processing (Xia et al., 2026b). In comparison, input feature selection has received less mechanistic consideration. The atmospheric physics governing farm-scale power production is increasingly well understood, yet has not been effectively translated into forecasting feature selection, particularly for vertical wind structure represented by commonly available multi-height observations.

Hub-height wind speed has long been the primary forecasting input (Mahoney et al., 2012; Ye et al., 2017; Che et al., 2022; Sun & Yang, 2023), reflecting a simplified representation of atmospheric inflow consistent with the single-turbine power curve, where power can be characterized by a single hub-height wind speed (Turbines—Part, 2005). This practice is further supported by the strong correlation between hub-height wind speed and single-turbine power (Clifton et al., 2013). At the wind farm scale, however, this simplification becomes insufficient because aggregate power depends not only on the inflow at individual turbines but also on wake interactions across the farm (Barthelmie et al., 2010). Wake development and recovery are strongly influenced by boundary-layer structure, including wind shear, turbulence, and atmospheric stability (Abkar & Porté-Agel, 2015). As a result, forecasting based solely on hub-height wind speed often fails to fully capture the atmospheric drivers of farm-scale power variability. Physical studies have demonstrated that rotor-equivalent wind speed (REWS) (Wagner et al., 2009), which integrates wind speeds across the rotor-swept layer, outperforms single-height measurements in forecasting farm-scale power (Sasser et al., 2022). Even with this advance, atmospheric stability can induce 15–20% power variations at the same nominal wind speed, and turbulent kinetic energy has been identified as a dominant modulation factor (Wharton & Lundquist, 2012). Confining vertical integration to the rotor-swept layer therefore remains insufficient to capture the influence of vertical structure and its governing scales on farm-scale power variability.

Flow-regime variability adds another dimension to farm-scale power representation, as wind farms operate under both free-stream and wake-influenced inflow conditions with distinct drivers of power production. Wake interactions can cause 10–20% power losses and elevate turbulence intensity (Barthelmie et al., 2007; Feroz et al., 2020), implying that the physical controls on power output vary dynamically with flow regime and atmospheric stability. Recent explainable-AI analyses also rank turbulence-related variables, including turbulent kinetic energy and turbulence intensity, among the most informative predictors of farm-scale power, second only to wind speed (Optis & Perr-Sauer, 2019; Kim & Kim, 2023), further suggesting regime-dependent modulation of inflow–power relationships. Yet data-driven forecasting models commonly incorporate multi-height meteorological inputs as homogeneous predictors, without explicitly accounting for flow-regime-dependent mechanisms. When distinct flow

regimes are aggregated, underlying physical differences can manifest as persistent empirical correlations that lack clear physical interpretation. One such example is the consistently reported high correlation between near-surface wind speeds (e.g., 10-m wind) and total wind-farm power output (Rubanenko et al., 2020; Chen & Xu, 2022), whose physical basis remains unclear.

Using multi-height observations from a flat-terrain wind farm, we link atmospheric flow physics to farm-scale power forecasting by partitioning inflow regimes and quantifying how vertical wind structure governs power variability under free-stream and wake-influenced conditions, thereby clarifying the physical origin of the persistent high correlation between near-surface winds and total farm power. We show that incorporating regime-dependent profile information outperforms both single hub-height input and undifferentiated multi-height predictors in observational power reconstruction and Numerical Weather Prediction (NWP)-driven day-ahead forecasting. By revealing that governing atmospheric time scales and vertical coherence underpin the relevance of non-hub-height winds to farm-scale power, the results provide a mechanistic basis for improving wind-to-power representation in operational forecasting.

2 Materials and Methods

2.1 Flow Regime Classification

Flow regimes were classified based on wind direction relative to the turbine layout (Figure S1 shows the turbine layout and mast location). Westerly winds (225° – 315°) are classified as free-stream conditions, under which measurements represent background atmospheric flow upstream of the turbine array. In contrast, easterly winds (45° – 135°) traverse the turbine array before reaching the mast and are therefore classified as wake-influenced conditions, under which wind profiles are affected by turbine-induced momentum deficits and enhanced turbulence (Porté-Agel et al., 2020). Regime classification is based solely on geometric wind-direction criteria relative to turbine layout and does not impose additional buffer zones near sector boundaries. To ensure high regime purity and exclude transitional or vertically sheared directional states, regime assignment required consistent wind direction across all four measurement heights (10 m, 30 m, 50 m, and 70 m). Over the study period, the resulting dataset comprises 1,951 free-stream samples (3.7%) and 8,693 wake-influenced samples (16.6%), with the remaining observations excluded from regime-specific evaluation.

2.2 Data Analysis Methods

Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN; Torres et al., 2011), an improved variant of EEMD (Wu & Huang, 2009) was applied to the continuous wind speed (at different heights) and total farm power time series to obtain intrinsic mode functions across temporal scales. Regime-dependent wind–power coupling was quantified by computing cross-scale correlations within subsets of timestamps classified into free-stream and wake-influenced conditions based on wind direction (Text S1). Empirical Orthogonal Function (EOF) analysis was performed on wind profiles to identify dominant modes of vertical structure variability (Wilson, 1996). The first EOF mode (EOF1) and its principal

component time series (PC1) capture the leading spatial pattern and its temporal amplitude (Wilks, 2011). Least-squares regression between PC1 and wind speeds at 10 m or 70 m quantifies how effectively each level captures vertical profile variability. Further details on the decomposition and EOF methodologies are provided in Texts S1 and S2.

2.3 Power Reconstruction Experiment

A power reconstruction experiment was conducted using a LightGBM regression model (Ke et al., 2017) trained on observed meteorological variables and farm-scale power output. Three input configurations were tested: hub-height wind speed only (HH, 70 m), Standard REWS integrating wind speeds within the rotor-swept layer (SR, 30–70 m), and Extended REWS incorporating surface-layer wind speed (ER, 10–70 m). Models were trained on a subset of the observational dataset and evaluated on withheld test samples. Performance was assessed on the full test set and separately on test samples under free-stream and wake-influenced conditions. Full implementation details, including model hyperparameters, data filtering, and train–test splitting procedures, are documented in Text S3. To examine whether the regression model exploits physically meaningful information rather than statistical artifacts, SHapley Additive exPlanations (SHAP) were applied to quantify the contribution of each input variable (Lundberg & Lee, 2017). SHAP analysis was conducted separately for free-stream and wake-influenced conditions to assess whether variable importance aligns with the regime-dependent vertical structure identified in Section 4.1.

2.4 Day-Ahead Forecast Experiment

To assess operational relevance, day-ahead forecasting experiments were conducted using meteorological variables predicted by the WRF model. Following common practice (Cassola & Burlando, 2012), predicted variables were bias-corrected against observations before being used for power prediction. The same three input configurations evaluated in the reconstruction experiment—hub-height wind speed only (HH), Standard REWS integrating wind speeds within the rotor-swept layer (SR), and Extended REWS including surface-layer wind speed (ER)—were applied to the bias-corrected WRF outputs. Temperature was included as a predictor in all configurations (Text S4).

In addition to these baseline configurations, a wind-direction-aware (WDA) forecasting strategy was implemented to reflect regime-dependent behavior. Forecasts were partitioned into free-stream conditions (225°–315°), wake-influenced conditions (45°–135°), and transitional directions (remaining sectors) based on 70-m wind direction predicted by WRF. During forecasting, the sector was determined from the predicted wind direction, and the corresponding sector-specific model was applied. The input variables for each sector were determined based on SHAP-derived feature importance from the reconstruction experiment (Section 4.3): for the free-stream sector, wind speeds at 10 m, 50 m, and 70 m were used; for the wake-influenced sector, wind speeds at 10 m, 30 m, and 70 m were used; for the transitional sector, wind speeds at all four heights were used. In all sectors, 10-m temperature was included as an additional predictor. Full implementation details are documented in Text S4.

All four forecasting strategies followed a two-stage learning framework. In the first stage, each meteorological variable was individually bias-corrected using a LightGBM model trained to map WRF outputs to the corresponding observed variable. In the second stage, the bias-corrected meteorological variables were used as inputs to a separately trained LightGBM power prediction model. For the HH, SR, and ER strategies, a single power model was trained on the full dataset using the respective input configurations. For the WDA strategy, three independent power models were trained separately within the free-stream, wake-influenced, and transitional sectors using sector-specific input configurations; during prediction, the sector was determined from the WRF-predicted 70-m wind direction, and the corresponding model was applied.

2.5 Statistical Testing

Model performance was evaluated using coefficient of determination (R^2), root mean square error (RMSE), and Spearman correlation coefficient computed against observed farm-scale power output. Statistical significance of performance differences between modeling strategies was assessed using the Diebold-Mariano test (Diebold & Mariano, 1995), which accounts for temporal autocorrelation in forecast errors when comparing predictive accuracy of competing models.

3 Data

3.1 Site and Observational Data

Observational data were collected from a 201 MW wind farm located on flat terrain in the desert Gobi of the Hexi Corridor, Gansu, China (96.8114°E, 40.2053°N). The farm comprises 134 wind turbines, each with a rated capacity of 1.5 MW, a hub height of 61.5 m, and a rotor diameter of 82 m. A 70-m meteorological mast is situated 586 m west of the farm's western boundary (Figure S1). Vertical wind profiles were characterized using wind speed measurements from cup anemometers at four heights (10 m, 30 m, 50 m, and 70 m), provided by the operator, from May 2021 to October 2022 at 15-minute intervals. In this study, the 10 m measurement represents the surface layer, while the 70 m level is used as a near-hub-height reference because it is the closest available measurement height to the turbine hub height.

All observational data underwent standardized quality-control consistent with Xia et al. (2026a; 2026b), including removal of missing, invalid, and physically implausible values (outside 0–30 m s⁻¹), interpolation of short gaps (≤ 2 consecutive time steps), and exclusion of statistical outliers exceeding three standard deviations from monthly means. Records were temporally synchronized with WRF outputs, and no additional filtering was applied to preserve realistic operational variability.

3.2 NWP Data

Day-ahead forecasting employed two WRF version 4.2 dynamical downscaling datasets: one driven by the ECMWF Integrated Forecasting System (EC-WRF) and one by the NCEP Global Forecast System (GFS-WRF), both using identical WRF configurations to ensure comparability. Following operational requirements, forecasts targeting Day+1 (00:00–23:45 Beijing Time) were

initialized daily at 20:00 Beijing Time, with the first 28 hours treated as spin-up. Forecast hours 28–52 were extracted, bilinearly interpolated to the mast location, and vertically interpolated to the four observation heights. Full WRF configuration and validation are documented in Xia et al. (2026a).

4 Results and Discussion

4.1. Observational Evidence: Vertical Correlation Structure and Flow-Regime Dependency

The meteorological mast, positioned 586 m west of the western boundary of the wind farm, captures two distinct inflow regimes determined by wind direction (Figure 1A): westerly winds represent free-stream conditions, whereas easterly winds correspond to wake-influenced conditions. This configuration provides a natural experimental setting for studying the relationship between vertical wind measurements and farm-scale power generation.

Contrary to conventional expectations, analysis of the full dataset shows that 10-m wind speed exhibits a slightly higher Spearman correlation with total farm power ($r \approx 0.81$) than hub-height wind speed at 70 m ($r \approx 0.79$) (Figure 1B, black line). This result challenges the conventional reliance on hub-height or rotor-layer wind speed for farm-scale power representation (Mahoney et al., 2012; Wagner et al., 2009; Ye et al., 2017). The complete pairwise Spearman correlation matrix, including inter-height correlations, is provided in Figure S2.

When partitioned by flow regime, this apparent anomaly is resolved. Under free-stream conditions (Figure 1B, red line), correlation increases with height and peaks at 70 m ($r \approx 0.83$), consistent with expectations based on vertical shear structure. Under wake-influenced conditions (Figure 1B, blue line), however, correlation at 70 m decreases substantially ($r \approx 0.70$), while correlation at 10 m remains high ($r \approx 0.78$). The higher overall correlation at 10 m therefore arises from degradation of the hub-height signal under wake-influenced conditions rather than from enhanced predictive power of near-surface wind.

This regime-dependent difference is also reflected in the vertical wind profiles (Figure 1C and 1D). Under free-stream conditions, wind speed increases smoothly from the surface to hub height, consistent with a typical shear structure. Under wake-influenced conditions, wind speeds between 30 m and 70 m are reduced and exhibit larger variability compared with free-stream flow. These changes reflect turbine-induced momentum extraction and enhanced turbulence (Barthelmie et al., 2007; Feroz et al., 2020). In contrast, the distribution of 10-m wind speed is nearly identical between free-stream and wake-influenced conditions, suggesting that this level is much less influenced by the wake than the above layer and continues to reflect the background atmospheric inflow.

Collectively, these observations (Figure 1) demonstrate that the higher correlation at 10 m arises from vertical inhomogeneity of wake propagation, with the 10-m level remaining outside cumulative wake effects and thus more representative of the background atmospheric state. This observational evidence provides justification for transitioning from single hub-height measurements to a multi-height representation that includes surface-layer measurements,

consistent with the recognition in rotor-equivalent wind speed studies that vertical profile information is essential for accurate power characterization (Wagner et al., 2009; Sasser et al., 2022). However, conventional REWS approaches integrate wind speeds only within the rotor-swept area (typically 30–70 m), neglecting the role of surface-layer observations. These findings motivate the use of an extended profile that incorporates 10-m measurements, recognizing that surface-layer wind can represent farm-scale power variability through vertical atmospheric coupling.

4.2. Scale-Dependent Coupling and Vertical Structure Consistency

Section 4.1 showed that 10-m wind speed remains representative of the background atmospheric state under wake-influenced conditions. Whether 10-m wind can also represent farm-scale power variability depends on the time scales at which power responds and on the degree of vertical coherence (quantified as the Spearman correlation) across heights. If farm-scale power is dominated by variability at time scales where 10-m and rotor-layer winds evolve coherently, then surface-layer wind may capture the relevant atmospheric signal.

Scale-resolved variance analysis (Figure 2A) distinguishes sub-daily wake-related variability from synoptic-scale atmospheric variability. At sub-daily periods, wind-speed variance at 70 m is notably higher under wake-influenced conditions than under free-stream conditions, reflecting additional variability introduced by turbine interactions. In contrast, at synoptic and longer periods, variance levels at 10 m and 70 m become nearly indistinguishable under both free-stream and wake-influenced conditions. This behavior indicates that low-frequency variability is governed primarily by large-scale atmospheric forcing, which exerts a relatively uniform influence across the vertical column (see Text S1 for a discussion of limitations at the longest periods).

Vertical coherence analysis (Figure 2B, upper panels) further supports this separation. At short time scales, coherence between 10 m and 70 m wind speeds is reduced under wake-influenced conditions. With increasing averaging period, coherence increases steadily and approaches similar values under both regimes at synoptic time scales. The correlation with power output (Figure 2B, lower panels) shows a consistent pattern: weak at hourly scales but increasing substantially toward one to three days under both regimes. These results indicate that farm-scale power variability is primarily associated with synoptic-scale atmospheric forcing rather than short-scale wake-driven fluctuations.

The large vertical separation between 10 m and the rotor-swept layer raises the question of whether the vertical structure remains coherent across regimes. EOF analysis provides this insight. The first EOF mode explains more than 93% of total vertical variance in all cases (Figure 2C), with nearly identical contributions under free-stream (93.2%) and wake-influenced (93.5%) conditions. The spatial pattern of this dominant mode remains nearly unchanged between regimes, indicating that wake effects modify the mean wind profile but exert limited influence on the primary mode of temporal variability. Under synoptic-scale forcing, the vertical wind profile evolves as a coherent structure governed by a single dominant mode in both free-stream and wake-influenced conditions.

The ability of surface-layer wind to scale profile intensity is examined through its regression with PC1 amplitude. Strong linear correlations are observed between 10-m wind speed and PC1, with nearly identical regression slopes under free-stream and wake-influenced conditions (0.65 and 0.66; Figure 2D). In contrast, the regression slope between 70-m wind speed and PC1 differs more substantially between regimes (0.54 to 0.62), indicating that wake effects alter the scaling capability of hub-height wind. The relationship between 10-m wind speed and column-integrated variability remains stable across regimes, even under wake-influenced conditions.

Taken together, these results show that farm-scale power variability is controlled primarily by synoptic-scale atmospheric forcing. Because 10-m wind speed varies coherently with this dominant mode and exhibits consistent scaling behavior across regimes, surface-layer observations can represent the atmospheric state that drives farm-scale power variability. These findings provide the physical basis for extending vertical wind representation beyond the rotor-swept layer.

4.3. Evaluation of the Extended Profile Representation via Power Reconstruction

Sections 4.1 and 4.2 established the physical basis for extending vertical wind representation beyond the rotor-swept layer. To test whether this representation improves farm-scale power prediction, a LightGBM model is used to reconstruct total power from the three input configurations (HH, SR, ER) described in Section 2.3.

Figure 3A shows that reconstruction accuracy improves systematically from HH ($R^2 = 0.61$) to SR ($R^2 = 0.64$) and further to ER ($R^2 = 0.72$). Including surface-layer wind therefore yields a clear improvement over models using only rotor-layer measurements. To assess robustness across flow regimes, the test dataset is partitioned into free-stream and wake-influenced conditions. Figure 3B presents regime-dependent reconstruction performance using models trained on the full dataset. Under free-stream conditions, accuracy increases from HH ($R^2 = 0.65$) to SR ($R^2 = 0.72$) and further to ER ($R^2 = 0.80$). Under wake-influenced conditions, overall performance decreases, but the relative ordering remains consistent. HH drops to $R^2 = 0.44$ and SR to 0.53, whereas ER maintains substantially higher skill with $R^2 = 0.71$. This pattern indicates that surface-layer wind compensates for the degraded rotor-layer signal under wake-influenced conditions.

To assess whether the relative importance of input variables is consistent with the physical mechanisms identified in Sections 4.1 and 4.2, separate models are trained for free-stream and wake-influenced conditions, and SHAP is applied to the corresponding test sets. Figure 3C reveals clear regime-dependent feature behavior. Under free-stream conditions, 10-m wind speed exhibits the largest SHAP contribution, followed closely by 70-m wind speed. Wind speeds at 30 m and 50 m contribute less and show similar magnitudes, indicating partial redundancy within the rotor-swept layer. This pattern suggests that while hub-height wind remains important, the contrast between surface-layer and hub-height wind speeds plays a central role in characterizing vertical profile intensity under free-stream conditions, consistent with the vertical coupling results in Section 4.2. Under wake-influenced conditions, feature importance shifts markedly. The contribution of 10-m wind speed increases further and

becomes dominant, whereas the contribution of 70-m wind speed decreases. Wind speeds within the rotor-swept layer show reduced influence relative to the surface layer. This shift is consistent with the findings in Sections 4.1 and 4.2.

In addition to wind speed, 10-m temperature ranks among the most influential predictors under both regimes, consistent with the density dependence of wind power generation. SHAP-derived feature importance aligns with the physically established relationships above, supporting its interpretability in diagnosing model behavior (Liao et al., 2024). Taken together, these results indicate that the performance gains associated with Extended REWS arise from physically meaningful atmospheric information rather than statistical artifacts, thereby supporting the validity of extending vertical wind representation beyond the rotor-swept layer.

4.4. Operational Implications: Wind-Direction-Aware Strategy

Section 4.3 demonstrated that extended vertical wind representation improves power reconstruction. In operational forecasting, however, NWP-provided inputs vary in reliability across flow conditions, suggesting that a fixed input set may not be optimal under all inflow regimes.

To reflect this regime-dependent behavior, a WDA forecasting strategy is implemented for day-ahead prediction. Forecasts are partitioned into free-stream, wake-influenced, and transitional sectors based on 70-m wind direction predicted by WRF. During forecasting, the sector is determined using the predicted wind direction, and the corresponding sector-specific model is applied.

Figures 4A and 4C present EC-WRF and GFS-WRF day-ahead forecast performance, respectively. For EC-WRF-driven forecasts, WDA achieves $R^2 = 0.37$ and $RMSE = 32.12$ MW, compared with $R^2 = 0.28$ ($RMSE = 34.31$ MW) for HH, $R^2 = 0.32$ ($RMSE = 33.42$ MW) for SR, and $R^2 = 0.35$ ($RMSE = 32.78$ MW) for ER. Similar improvements are observed for GFS-WRF-driven forecasts, where R^2 increases from 0.24 for HH to 0.35 for WDA, with SR and ER yielding intermediate performance. By adopting sector-specific input configurations informed by SHAP analysis—using three wind speed levels for free-stream and wake-influenced sectors while retaining all four levels for transitional sectors—WDA reduces redundant and wake-contaminated predictors and outperforms ER.

Regime-resolved results (Figure 4B and 4D) clarify the source of improvement. Under wake-influenced conditions, the correlation between predicted and observed power increases from 0.59 (HH) to 0.66 (WDA) for EC-WRF-driven forecasts and from 0.59 to 0.63 for GFS-WRF-driven forecasts. In contrast, differences among configurations are relatively small under free-stream conditions. This is consistent with earlier findings on wake-related hub-height signal degradation. Allowing input usage to vary with flow regime therefore yields the largest benefit under wake-influenced conditions. These results demonstrate that the physical understanding of scale-dependent vertical coupling can be translated into a simple operational strategy. This approach complements recent efforts to account for wake effects in wind power post-processing (Romero-Barrera et al., 2026).

Several limitations should be noted. First, the analysis is based on a single wind farm in flat terrain, where vertical coupling between surface and rotor layers is relatively strong. The applicability of this strategy to complex terrain requires further multi-site evaluation. Second, the observational dataset is limited to mast measurements from 10 to 70 m, extending up to hub height but not above it. As a result, the role of wind structure in the upper part of the rotor-swept layer remains unexamined. This aspect warrants future investigation using scanning or profiling lidars, which can provide continuous wind profiles above hub height and across the full rotor layer. Third, implementation depends on the availability of multi-height observations and reliable wind-direction forecasts, which may vary across operational settings.

5 Conclusions

This study finds that 10-m surface-layer wind speed maintains a stronger correlation with farm-scale power than hub-height wind speed under wake-influenced conditions, explains the physical mechanisms underlying this observation, and proposes an extended profile forecasting approach grounded in this understanding. While REWS approaches recognize the importance of vertical structure, this study establishes that the relevant profile must extend below the rotor-swept layer under certain flow conditions.

Regime-based partitioning is essential for understanding this vertical dependence. Under free-stream conditions, measurements at 10 m and 70 m provide complementary information for characterizing vertical shear and profile development, especially when wind speeds within the rotor-swept layer exhibit redundancy. Under wake-influenced conditions, the 10-m level remains largely unaffected by wakes and serves as a key indicator of overall inflow profile intensity. Scale-resolved analysis reveals that at the synoptic timescales governing power variability, wind speeds vary coherently across the vertical column over flat terrain. EOF analysis shows that a single mode captures over 93% of vertical variance, enabling 10-m wind to serve as a consistent scaling factor for inflow profile intensity. Through this mechanism, near-surface observations provide a reliable reference for the atmospheric state driving power potential, even when rotor-layer measurements are degraded by wake effects.

Building on this physical understanding, the extended profile representation integrates surface-layer and rotor-layer wind information to improve power reconstruction accuracy. When further combined with regime-specific feature selection, a wind-direction-aware strategy yields systematic improvements in NWP-driven day-ahead forecasts. These results indicate that input selection should follow governing atmospheric scales and flow regimes, not solely measurement height proximity to the rotor. Recognizing and exploiting vertical coupling between surface and rotor layers provides a physically grounded pathway for improving farm-scale NWP-driven wind power forecasting.

Open Research

The observational wind speed and power data, together with the EC-WRF and GFS-WRF dynamical downscaling outputs used in this study, are available with restricted access at Science Data Bank (<https://doi.org/10.57760/sciencedb.37695>). The analysis and visualization code is publicly available at Zenodo (<https://doi.org/10.5281/zenodo.18671960>) and GitHub

(<https://github.com/Xiax24/ExtendedProfile-WindPower-Forecasting>). ECMWF IFS HRES forecasts are publicly available from <https://www.ecmwf.int/en/forecasts/datasets/set-i>. NCEP GFS forecasts are publicly accessible from <https://www.ncei.noaa.gov/products/weather-climate-models/global-forecast>. The WRF model version 4.2 is openly accessible from https://www2.mmm.ucar.edu/wrf/users/download/get_source.html. The LightGBM package used for machine learning experiments is available from <https://github.com/microsoft/LightGBM>.

Conflict of Interest Disclosure

The authors declare there are no conflicts of interest for this manuscript.

Author contribution

Xin Xia: Writing – original draft, Writing – review & editing, Conceptualization, Data curation, Formal analysis, Methodology, Visualization. **Yong Luo:** Writing – review & editing, Data curation, Resources, Validation, Supervision. **Rui Chang:** Validation, Data Curation. **Martin Dörenkämper:** Writing – review & editing, Validation. **Rong Zhu:** Conceptualization, Validation. **Kun Yang:** Writing – review & editing, Validation. **Peidu Li:** Software, Visualization. **Xiaoli Zhang:** Validation.

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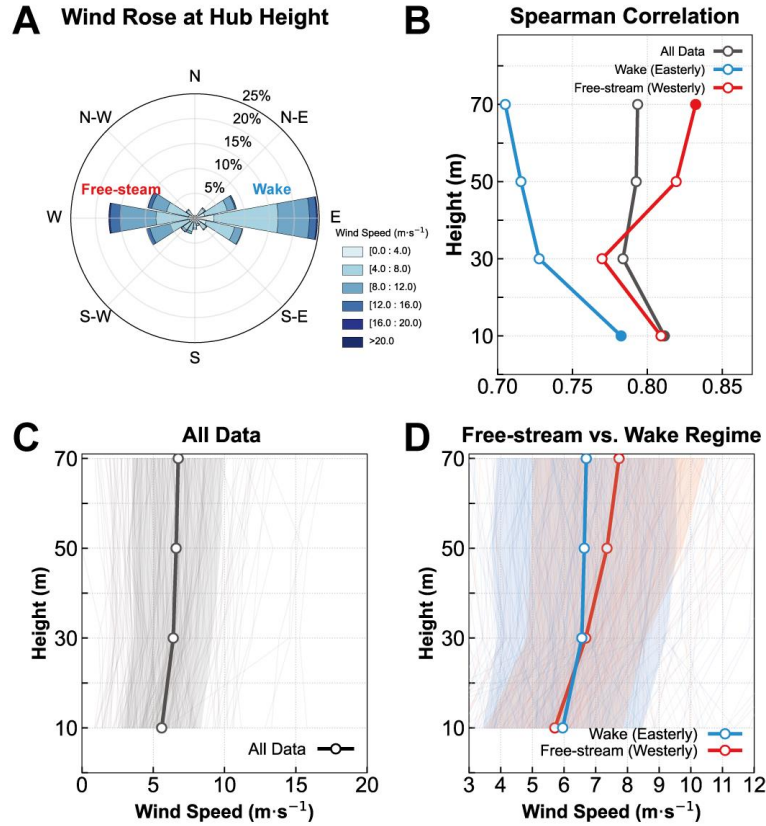


Figure 1. Regime-dependent vertical structure of wind speed and correlation with power production. A: Wind rose at 70-m hub height. The meteorological mast location 586 m west of the first turbine column (33 turbines) enables categorization into westerly free-stream and easterly wake effect sectors. B: Vertical profiles of Spearman correlation between height-specific wind speed and total power output for aggregate dataset (black), free-stream sector (red), and wake effect sector (blue). C: Mean vertical wind speed profile for the entire observation period with background thin lines representing individual instantaneous profiles and gray shading representing one standard deviation ($\pm 1\sigma$). D: Comparison of mean vertical wind speed profiles between free-stream (red) and wake effect (blue) regimes with background thin lines showing individual profiles for each regime. Shaded areas denote one standard deviation ($\pm 1\sigma$).

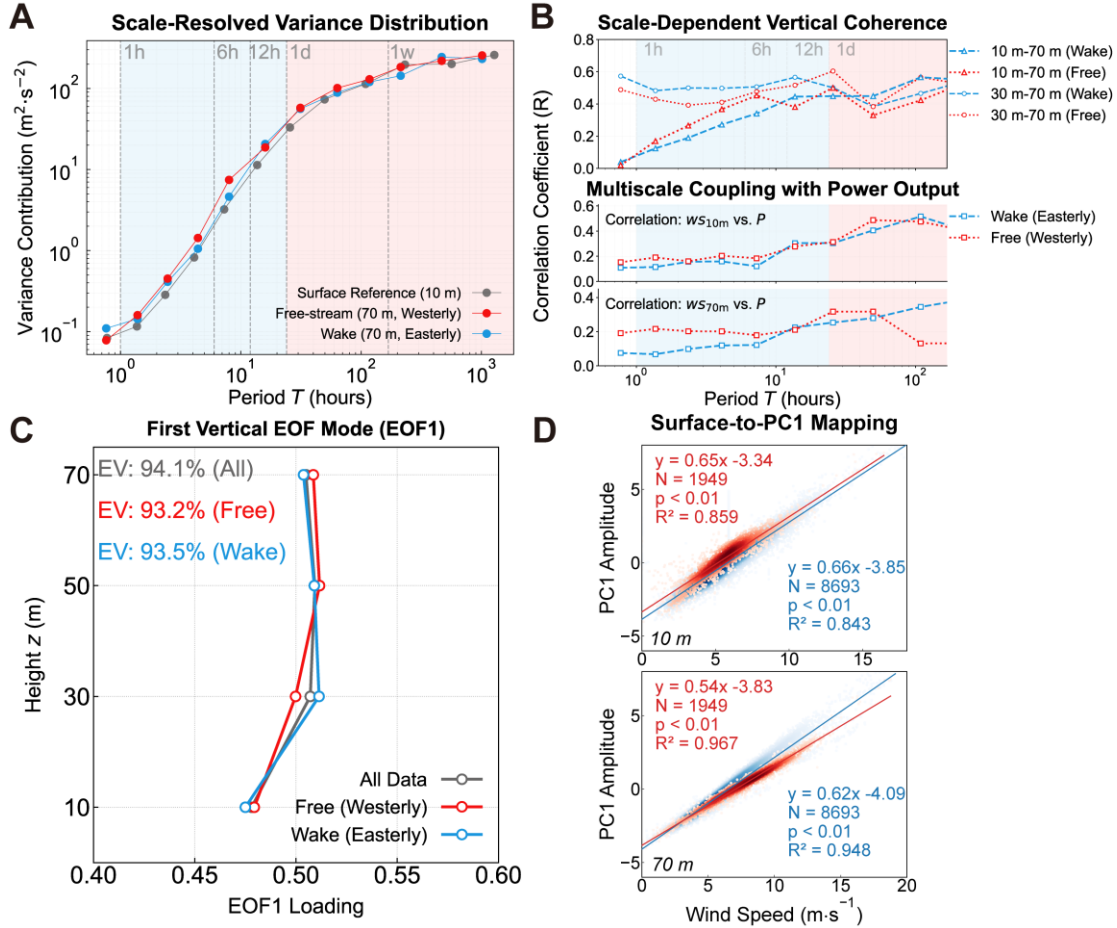


Figure 2. Scale-resolved variance distribution, vertical coherence, and linear relationship between surface wind and vertical profile energy state. A: Scale-resolved variance contribution of wind speed at 10 m (gray) and 70 m for free-stream (red) and wake effect (blue) conditions. Vertical dashed lines indicate temporal periods of 1h, 6h, 12h, 1d, and 1w. B: Scale-dependent vertical coherence and correlation with power output. Top panel shows Spearman correlation coefficient between wind speeds at different height pairs (10-70 m, 30-70 m) as a function of time scale. Bottom panels show Spearman correlation between power output and wind speed at 10 m and 70 m across multiple time scales. Shaded regions differentiate high-frequency and synoptic-scale regimes. C: Vertical profiles of EOF1 loadings for aggregate dataset (gray), free-stream (red), and wake effect (blue) regimes. Explained variance (EV) for each regime is indicated. D: Linear regression between 10-m wind speed (top) or 70-m wind speed (bottom) and PC1 amplitude of the vertical wind profile. Regression statistics including coefficient of determination, sample size, and significance level are shown for free-stream (red) and wake effect (blue) conditions.

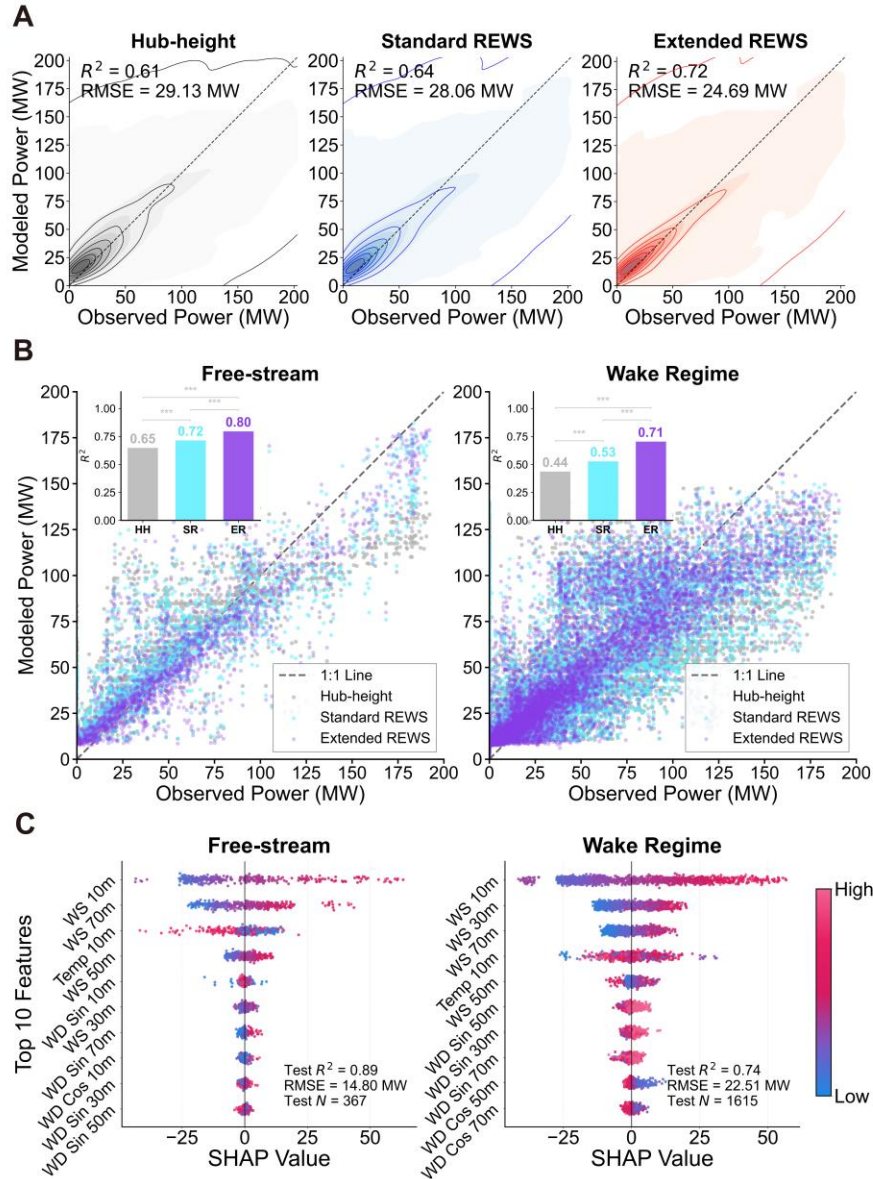


Figure 3. Comparative performance and feature importance analysis of power reconstruction frameworks. A: Density scatter plots comparing observed power with power reconstructed by LightGBM using three input configurations: Hub-height (HH), Standard REWS (SR), and Extended REWS (ER). Coefficient of determination and root-mean-square error are shown for each framework. B: Performance comparison across flow regimes showing coefficient of determination for the three frameworks under free-stream and wake effect conditions. Bar chart insets show significant performance improvement of ER over SR and HH based on Diebold-Mariano tests ($***p < 0.01$). C: SHAP analysis for the ER model under free-stream and wake effect regimes. Features are ranked by mean absolute SHAP value representing relative importance. Each point represents an individual sample with color indicating feature value magnitude and horizontal position showing impact on power prediction.

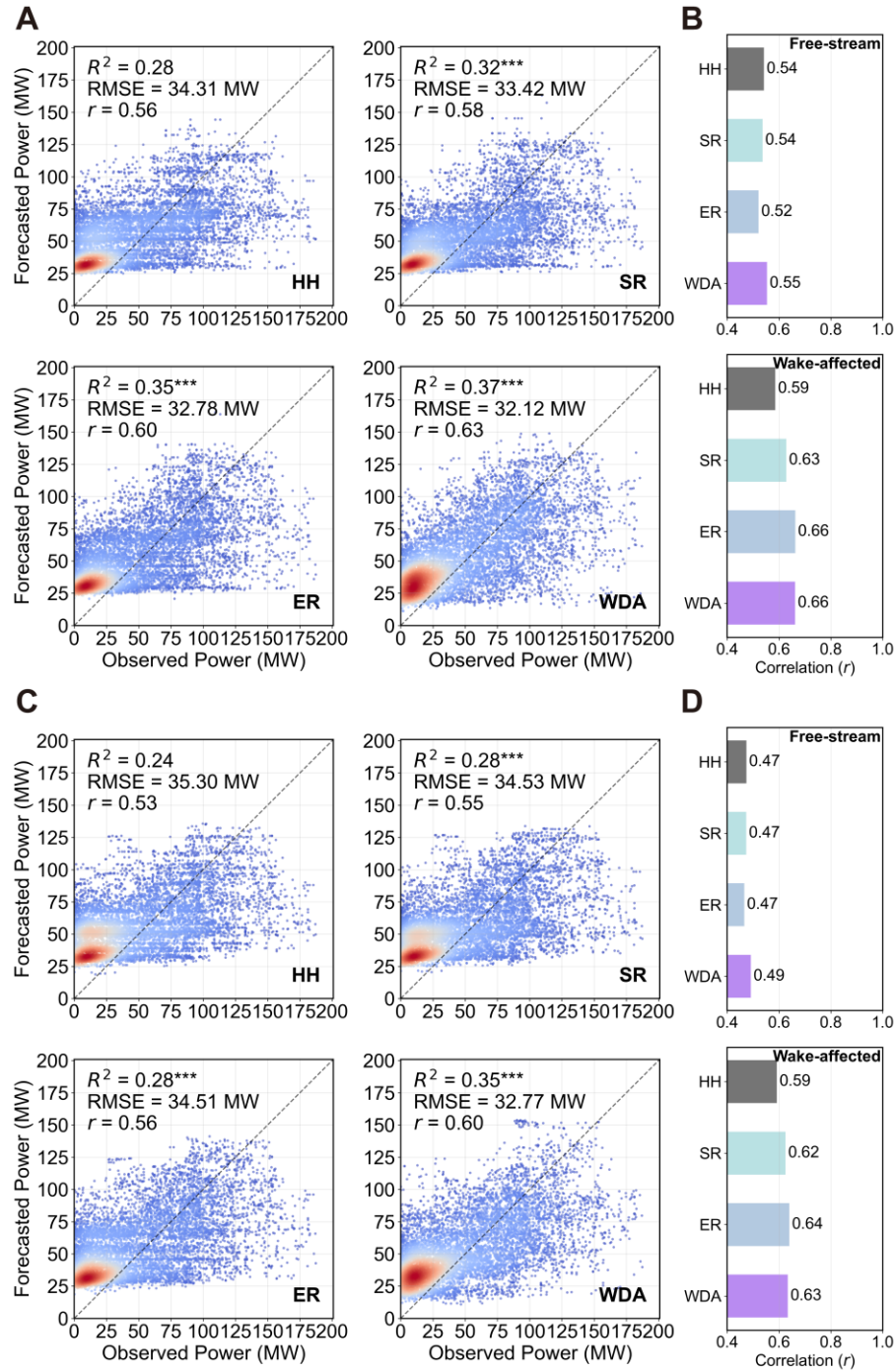


Figure 4. Operational evaluation of WDA strategy for day-ahead power forecasting. A and C: Density scatter plots comparing observed and forecasted power using four configurations: Hub-height (HH), Standard REWS (SR), Extended REWS (ER), and Wind-Direction-Aware (WDA). Results shown for A: EC-WRF and C: GFS-WRF driven forecasts on complete test dataset. Coefficient of determination, root-mean-square error, and correlation coefficient are shown for each configuration. Asterisks (***) indicate statistical significance at $p < 0.01$ level based on

586 Diebold-Mariano tests relative to HH baseline. B and D: Correlation coefficients across free-
587 stream and wake effect regimes for B: EC-WRF and D: GFS-WRF.
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